Math 220: Mathematical Reasoning and Proof
Instructor: David Zureick-Brown ("DZB")

All assignments

Last updated: October 30, 2026

Gradescope code: 8D24GR

Show all work for full credit!
Proofs should be written in full sentences whenever possible.

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Gradescope Instructions for submitting work in Math 220

You will be using the online Gradescope progam to submit your homework and exams. These instructions tell you how to sign up initially, and how to submit your written work.

Signing up for Gradescope the first time.

If you haven't used Gradescope for an **Amherst College** course before:

- Go to `http://www.gradescope.com`, click on "Sign up for free" (which may auto-scroll you to the bottom of the page), and select Sign up as [a] "Student".
- In the signup box:
  - Use the course entry code **8D24GR**
  - Use your full name
  - Use your **Amherst College email** address. Or, if you are a Five-College student, use your email address from your own school.
  - Leave the "Student ID" entry blank.
- You will probably get an email asking to set a password for your account, so check your `amherst.edu` email inbox. (Or your email inbox through your own school, for Five-College students.)

Adding Math 220 to Gradescope.

If you **have** used Gradescope for an Amherst course before, and so you already have an account through your `amherst.edu` email, you still need to add Math 220, so:

- Go to `http://www.gradescope.com` and log in.
- Go to your Account Dashboard (click the Gradescope logo at upper left), and click "Add Course" at bottom right.
- Use the course code **8D24GR**

(submission instructions on next page)

Submitting written work

First write it out on paper as you would normally. Then **scan it** to create a PDF. One method for scanning is the smartphone app **DropBox**. It makes nice clear scans, and it saves them directly into a folder so that you can have all your assignments in one place. **CamScanner** is another free scanning App, and there are others, too. **Gradescope** now has its own scanning app. You can also use a printer/scanner if you prefer.

Please be kind to our dear graders and make sure your submission is legible !

In particular, please leave some spacing between separate problems.

If you have a tablet computer, you may write your work there (instead of on paper) and save it as a PDF.

Some of you may know the math formatting package LaTeX and may want to use it in Math 220. That's fine, too; if so, you may write up your work in LaTeX and save the resulting PDF.

In short, any method is fine as long as it creates a legible **PDF** file and **NOT a photo**.

For example, if you use the DropBox app, then in your created *Math 220 Homework* Dropbox folder, you can select create (+) at the bottom of the screen and click the *Scan Document* option. Snap a shot of the first page of your homework, and then click [+] to snap shots of any subsequent pages. Do **not** use the *Take Photo* option.

After you have scanned/saved your work as a PDF, submit it on Gradescope as follows:

- Go to `http://www.gradescope.com` and log in.
- Select the course "Math 220, Fall 2026" and the appropriate assignment.
- Select "submit pdf" to submit your work in PDF format. Browse to find your PDF and upload.
- Now it is time to **tag** your problems. This is an **important step**, where you are telling Gradescope which problems are on which page(s).

  For each problem, select the pages of your submission where your written solution appears.

  I think the easiest thing to do is to click on the page of **your** homework upload where you wrote the given problem, and then click on the assigned problem listed. Repeat for each problem.

  You must tag the problems or else you will not get credit for your work.

  Gradescope will give you a warning when you go to submit your assignment if you have not selected the pages correctly. If you tag a problem incorrectly, you can fix it by clicking "More" and "Reselect Pages".
- Click Save or Submit.

After your assignment is graded, you will be able to see your score on the written problems, along with comments, on Gradescope. You should receive an email notifying you when each homework set is graded.

Assignment 1: Introduction to course. First proof

Due by 10:00am, eastern, on Friday, September 4

Suggested readings for this problem set:

- Syllabus: https://dmzb.github.io/teaching/2026Fall220/syllabus-F26-220.pdf
- Gradescope instructions (previous page)
- No reading assigned, but good to start reading Chapter 1

All readings are from Bonde and Keane, *Introduction to Abstract Mathematics*.

Assignment: due Friday, September 4, 10:00am, via Gradescope (8D24GR):

Please remember that solutions must be handwritten.

1. Using the definitions of "divides" and "even", explain why 0 is even. Explain why 7 divides 0, but 0 does not divide 7.
2. Suppose that n is an even integer, and let m be any integer. Prove that nm is even.
3. Suppose that n and m are odd integers. Prove that $n + m$ is an even integer. (Hint: an integer n is odd if and only if there exists an integer k such that $n = 2k + 1$.)
4. After reading the syllabus, write out the statment: "I read the syllabus and am aware of this course's policies".
5. Recall that the homework is graded only for completion. If there are particular problems you would like grader feedback on, please list them here.

Remember to use only the definitions in your proofs!

In particular, do not use phrases like "multiple of" or "is a factor"

Assignment 2: Logic. "Direct" proofs

Suggested readings for this problem set: **Finish reading chapter 1**
All readings are from Bonde and Keane, *Introduction to Abstract Mathematics*.

Assignment: due Friday, September 11, 10:00am, via Gradescope (8D24GR):

1. Write the negation of each of the following statements.

   (a) All triangles are isosceles.
   (b) Every door in the building was locked.
   (c) Some even numbers are multiples of three.
   (d) Every real number is less than 100.
   (e) Every integer is positive or negative.
   (f) If f is a polynomial function, then f is continuous at 0.
   (g) If $x^2 > 0$, then $x > 0$.
   (h) There exists a $y \in \mathbf{R}$ such that $xy = 1$.
   (i) $(2 > 1)$ and $(\forall x, x^2 > 0)$
   (j) $\forall \epsilon > 0$, $\exists \delta > 0$ such that if $|x| < \delta$, then $|f(x)| < \epsilon$.

2. Write the converse, contrapositive, and negation of each of the following implications.

   (a) If a quadrilateral is a rectangle, then it has two pairs of parallel sides.
   (b) $(P \wedge \neg Q) \Rightarrow R$
   (c) $P \Rightarrow (R \Rightarrow \forall x, Q(x))$

3. Let P and Q be statements. Write the truth table for

   (a) $(\neg P) \vee Q$
   (b) $((\neg P \Rightarrow Q) \wedge \neg Q) \Rightarrow P$

4. Are the statements $(P \vee Q) \wedge R$ and $P \vee (Q \wedge R)$ equivalent? If so, give a proof. If not, explain why by giving a counterexample.

   (Two statement forms are equivalent if they have the same truth tables, and here, a counterexample simply means some choice of truth values for P, Q, and R such that the two statement forms give different outputs.)

5. Let P and Q be statements.

   (a) Prove that $\neg(P \Rightarrow Q)$ is equivalent to $P \wedge \neg Q$.
   (b) Prove that $\neg(P \Rightarrow Q)$ is *not* equivalent to $\neg P \wedge Q$.
   (c) Give an example of statements P and Q such that $\neg P \Rightarrow \neg Q$ is true and $\neg(P \Rightarrow Q)$ is false.

6. Suppose that n is an odd integer. Prove that n^2 is an odd integer. (Hint: an integer n is odd if and only if there exists an integer k such that $n = 2k + 1$.)

7. Prove that if n^2 is even, then n is even. (Click here for a hint. Also: you are allowed to use the results of previous problems.)

8. Prove that m is even if and only if $m^2 + 6$ is even.

   (For an "if and only if" proof, you must do "two" proofs; for this problem this means that you need to prove the statement "if m is even then $m^2 + 6$ is even" and also the statement "if $m^2 + 6$ is even then m is even". A hint for the second statement is that you can also use the previous problem in your proof.)

9. Recall that the homework is graded only for completion. If there are particular problems you would like grader feedback on, please list them here.

Assignment 3: More "Direct" proofs; proof by cases

Suggested readings for this problem set: **Section 5.4**
All readings are from Bonde and Keane, *Introduction to Abstract Mathematics*.

Assignment: due Friday, September 18, 10:00am, via Gradescope (8D24GR):

1. Prove that for all $a \in \mathbb{Z}$ and for $n \in \mathbb{Z}_{\geq 0}$, $a - 1$ divides $a^n - 1$.

2. Suppose that $a \mid b$. Prove that for all $n \in \mathbb{Z}_{>0}$, $a^n \mid b^n$.

3. Suppose that there exists an integer $n \in \mathbb{Z}_{>0}$ such that $a \mid b^n$. Is it true that $a \mid b$? Prove or disprove your answer. (For a disproof, please give a counterexample that demonstrates that the statement is false.)

4. Prove that for all integers n, n and $n + 1$ have no common divisors other than ± 1.

   (A *common divisor* of a and b is an integer d that divides both a and b. So prove that if d is a common divisor of n and $n + 1$, then d is either 1 or -1.)

5. Prove that the product of three consecutive integers is divisible by 6.

   (It suffices to prove that it is divisible by 2 and 3 separately.)

6. Show that for all integers a and b,
   $$a^2b^2(a^2 - b^2)$$
   is divisible by 12.

   (It suffices to prove that it is divisible by 4 and 3 separately.)

7. Finish the following proofs from class.

   (a) In class, when we proved that $|xy| = |x|\,|y|$, we only did the cases when $x \geq 0$. Prove that $|xy| = |x|\,|y|$ when $x < 0$. (There are two cases for y).

   (b) In class, when we proved that $|x + y| \leq |x| + |y|$, we only did the cases when $x, y \geq 0$, when $x, y < 0$, and when $x \geq 0, y < 0$. Prove that $|x + y| \leq |x| + |y|$ when $x < 0, y \geq 0$

8. Consider the following statements about inequalities.

   (a) Prove that $|a - b| < c$ if and only if $b - c < a < b + c$. (You should only need the definition of absolute value for this one.)

   (b) Prove or disprove: if $a - b < c$, then $|a - b| < c$.

   (c) Prove or disprove: if $|a - b| < c$, then $a - b < c$.

   (d) Prove that $\big||x| - |y|\big| \leq |x + y|$. (You can use a fact from Wednesday's class for this one.)

   (For a disproof, give a **counterexample**, i.e., an example of a, b, c demonstrating that the statement is false. You are also allowed to use previous problems and statements from class.)

9. Recall that the homework is graded only for completion. If there are particular problems you would like grader feedback on, please list them here.

Assignment 4: Proof by contradiction

Due by 10:00am, eastern, on Friday, September 25

Suggested readings for this problem set: **Section 1.4, p. 41-42 (stop at Historical Comments)**
All readings are from Bonde and Keane, *Introduction to Abstract Mathematics*.

Assignment: due Friday, September 25, 10:00am, via Gradescope (8D24GR):

1. Prove that there do not exist integers a, and b such that $21a + 30b = 1$.
2. Find all positive integers n such that $n^2 - 1$ is prime. Prove that your answer is correct.
3. Prove that if x is an integer, then $x^2 + 2$ is not divisible by 4.

   (Hint: there are two cases: x is even, x is odd. Also, feel free to use basic facts about even or odd, e.g., "odd + odd = even", without additional proof.)
4. Prove that there are no integer solutions to the equation $x^2 - y^2 = 2$.
5. Let a, b, c be integers satisfying $a^2 + b^2 = c^2$. Prove that at least one of a, b, or c is even.
6. Suppose that x is a real number such that $0 \leq x \leq \pi/2$. Prove that $\sin x + \cos x \geq 1$.

   (Hint: "square both sides" and at some point in your proof, use that $(\sin x)^2 + (\cos x)^2 = 1$.)
7. Prove that $2^{1/3}$ is irrational.
8. Suppose that a and n are integers that are both at least 2. Suppose that $a^n - 1$ is prime.

   (a) Prove that $a = 2$.

   (b) Prove that n is a prime.

Recall that the homework is graded only for completion. If there are particular problems you would like grader feedback on, ~~please list them here.~~ Put a star by your problem number. (E.g., if you want feedback on problem 3, put a star by the number 3 in your solution.)

Assignment 5: Induction

Due by 10:00am, eastern, on Friday, October 2

Suggested readings for this problem set: **Section 5.2, p. 159-163**
All readings are from Bonde and Keane, *Introduction to Abstract Mathematics*.

Assignment: due Friday, October 2, 10:00am, via Gradescope (8D24GR):

1. Prove that for every positive integer n,
$$1^3 + 2^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}.$$
2. Let a_n be defined recursively by $a_1 = 1$ and $a_n = \sqrt{1 + a_{n-1}}$. Prove that for all positive integers n, $a_n < 2$.
3. Let $F_1, F_2, F_3, \ldots = 1, 1, 2, 3, 5, 8, \ldots$ be the Fibonacci sequence. Prove that $F_1^2 + \cdots + F_n^2 = F_n F_{n+1}$.
4. Prove that $n! > 2^n$ for all $n \geq 4$.
5. Prove (using induction) that for all integers $n \geq 1$, $2^{2n} - 1$ is divisible by 3.
6. *Bernoulli's inequality*: let $\beta \in \mathbb{R}$ be a real number such that $\beta > -1$ and $\beta \neq 0$. Prove that for all integers $n \geq 2$, $(1+\beta)^n > 1 + n\beta$.
7. Prove that for all integers $n \geq 1$,
$$1 + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} \geq \sqrt{n}.$$
8. Prove by induction that if $x_1, x_2, \ldots, x_n$ are real numbers, then
$$|x_1 + x_2 + \cdots + x_n| \leq |x_1| + \cdots + |x_n|.$$
(Note that there are two base cases, and note that if we have already proved a base case, we don't need to reprove that case.)

Recall that the homework is graded only for completion. If there are particular problems you would like grader feedback on, ~~please list them here.~~ Put a star by your problem number. (E.g., if you want feedback on problem 3, put a star by the number 3 in your solution.)

Midterm 1 study guide

In class on Wednesday October 7

Content: The questions will all be either

1. homework problems,
2. suggested problems,
3. problems we worked in class, or
4. minor variations of one of these.

Problems with very long proofs or that involved some unusual trick will not be on the exam.

You are allowed to use any previous problem from class or from the homework (e.g., "additivity of divisibility" or "the 2 out of 3 rule") on the exam without reproving it, unless otherwise noted on the exam. (E.g., if I ask you to prove "additivity of divisibility" on the exam, you will need to prove this using only the definition of divisibility, and I will remind you of this in the statement of the problem.)

A typical exam will have one or two questions from each week of the course, and this exam will cover **assignments 1-5**. You can expect problems about following:

- Negations
- Give definitions (e.g., divides, rational)
- Boolean Logic
- Direct proofs
- Proof by contrapositive
- Divisibility problems
- Contradiction
- Induction
- True/False problems.

There will be a negation problem, at least one definition, and around 4-6 problems involving proofs (possibly including "prove or disprove" problems).

For definitions, I want a definition, in prose (complete sentences), and I want "just" the definition, and not any additional facts about the definition. (E.g., if you give the definition of rational, do not include that a rational number can be written in reduced form; that is a fact about rational numbers not part of the definition of rational.)

Assignment 6: Set theory. Basic operations. Proofs with sets

Due by 10:00am, eastern, on Monday, October 19

Suggested readings for this problem set: **Section 2.1, p. 49-57; Section 2.2, p. 61-65 (stop at DeMorgan's laws)**
All readings are from Bonde and Keane, *Introduction to Abstract Mathematics*.

Assignment: due Monday, October 19, 10:00am, via Gradescope (8D24GR):

1. Let $A = \{0, 1, 2\}$. Let $B = \{1, \{2\}, 3\}$. Which of the following statements are true? (No justification is needed.)

   (a) $0 \in A$
   (b) $\{0\} \subseteq A$
   (c) $\{0\} \in A$
   (d) $A = B$
   (e) $2 \in B$
   (f) $\{2\} \in B$
   (g) $\{2\} \subseteq B$
   (h) $2 \in A \cup B$
   (i) $2 \in A \cap B$
   (j) $2 \in A - B$

2. Let $A = \{n \in \mathbb{Z} \mid n \text{ is a multiple of } 4\}$ and $B = \{n \in \mathbb{Z} \mid n^2 \text{ is a multiple of } 4\}$

   (a) Prove or disprove: $A \subseteq B$.
   (b) Prove or disprove: $B \subseteq A$.

3. Recall that $(a, b) = \{x : x \in \mathbb{R} \mid a < x < b\}$. Prove or disprove each of the following:

   (a) $(-1, 1) \subseteq (-2, 2)$.
   (b) $(-1, 2) \subseteq (-2, 1)$.

4. Prove that $(-10, 5] \cap [0, 10] = [0, 5]$.

5. Prove that if $A \not\subseteq C$ then $A \not\subseteq B$ or $B \not\subseteq C$. (Remember that you can reference and use lemmas and other things we proved in class.)

6. Let A, B be sets. Prove each of the following:

   (a) $A \cap B \subseteq A$;
   (b) $A \cap \emptyset = \emptyset$;

7. Prove that $A \cup (A \cap B) = A$.

8. Let n and m be integers. Prove that if m divides n then $n\mathbb{Z} \subseteq m\mathbb{Z}$.

Recall that the homework is graded only for completion. If there are particular problems you would like grader feedback on, ~~please list them here.~~ Put a star by your problem number. (E.g., if you want feedback on problem 3, put a star by the number 3 in your solution.)

Assignment 7: More sets. DeMorgan's laws. Cartesian Products. Power sets

Due by 10:00am, eastern, on Friday, October 23

Suggested readings for this problem set: **Section 2.2, p. 65-66; Section 2.3, p. 72, just the part about power sets.**
All readings are from Bonde and Keane, *Introduction to Abstract Mathematics*.

Assignment: due Friday, October 23, 10:00am, via Gradescope (8D24GR):

1. Let A, B and C be sets.
   (a) Prove that $(A \subseteq C) \wedge (B \subseteq C) \Rightarrow A \cup B \subseteq C$.
   (b) State the contrapositive of part (a).
   (c) State the converse of part (a). Prove or disprove it.
2. Let A, B, and C be sets. Prove or disprove the following. (For a disproof, please give an explicit counterexample; i.e., give an example of sets A, B and C demonstrating that the statement is false.)
   (a) If $A \not\subseteq B$ and $B \not\subseteq C$, then $A \not\subseteq C$.
   (b) If $A \subseteq B$ and $B \not\subseteq C$, then $A \not\subseteq C$.
3. Let A, B, C be sets. Prove each of the following:
   (a) Suppose that $B \subseteq C$. Prove that $A - C \subseteq A - B$.
   (b) $A \subseteq B$ if and only if $A \cap B = A$.
4. Let A, B and C be sets. Prove or disprove each of the following. (For a disproof, please give an explicit counterexample; i.e., give an example of sets A, B and C demonstrating that the statement is false.)
   (a) $(A \cap B) \cup C = A \cap (B \cup C)$.
   (b) $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$.
   (c) $(A \cup B) \cap \overline{A} = B - A$.
5. Prove that $c \in (a - b, a + b)$ if and only if $|a - c| < b$. (Recall that you are allowed (and encouraged) to use the results of previous homework problems without reproving them.)
6. Let A and B be sets. Prove that if $A \subseteq B$, then $P(A) \subseteq P(B)$. State the converse of this and prove or disprove it.

MORE PROBLEMS ON THE NEXT PAGE →

7. Let $A = \{0, 1, 2\}$. Which of the following statements are true? (No justification is needed.)

   (a) $0 \in P(A)$;

   (b) $\{0\} \in P(A)$;

   (c) $\{0\} \subseteq P(A)$;

   (d) $\{1, 2\} \in P(A)$;

   (e) $\{1, \{1\}\} \subseteq P(A)$.

   (f) $\{\{0, 1\}, \{1\}\} \subseteq P(A)$;

   (g) $\emptyset \in P(A)$;

   (h) $\emptyset \subseteq P(A)$;

   (i) $\{\emptyset\} \in P(A)$.

   (j) $\{\emptyset\} \subseteq P(A)$;

8. (a) For $i \geq 2$, let $A_i = (1/i, i] = \{x \in \mathbb{R} \mid 1/i < x \leq i\}$. Compute

$$\bigcup_{i=2}^{\infty} A_i \qquad \text{and} \qquad \bigcap_{i=2}^{\infty} A_i.$$

   (By "compute", I mean write these as a single set, using interval notation.)

   (b) For $i \geq 1$, let $A_i = [1, 1 + 1/i]$. Prove that

$$\bigcup_{i=1}^{\infty} A_i = [1, 2] \qquad \text{and} \qquad \bigcap_{i=1}^{\infty} A_i = \{1\}.$$

Recall that the homework is graded only for completion. If there are particular problems you would like grader feedback on, ~~please list them here.~~ Put a star by your problem number. (E.g., if you want feedback on problem 3, put a star by the number 3 in your solution.)

Assignment 8: Introduction to functions; images and surjectivity

Due by 10:00am, eastern, on Friday, October 30

Suggested readings for this problem set:

Section 3.1, p. 81-90 (stop at "Inverse Image"), Section 3.2, p. 97-100 (stop at Injective Functions)

All readings are from Bonde and Keane, *Introduction to Abstract Mathematics*.

Assignment: due Friday, October 30, 10:00am, via Gradescope (8D24GR):

1. Let $f\colon A \to B$ be a function. Finish the following sentence: an element $b \in B$ is not in the image of f if $\ldots$
2. Let $f\colon \mathbf{R} \to \mathbf{R}$ be the function defined by $f(x) = 6x + 5$.
   (a) Prove that $f(\mathbf{R}) = \mathbf{R}$.
   (b) Compute $f([1, 4])$. Prove your answer.
3. Let g be the function $g\colon \mathbf{Z} \to \mathbf{Z}$, where $g(x) = 2x + 1$.
   (a) Compute the image of g. Prove that your answer is correct.
4. Let $f\colon \mathbf{R} \to \mathbf{R}$ be the function defined by $x^4 + x^2$.
   (a) Compute the image of f. Prove that your answer is correct.
   (b) Compute $f([-1, 2])$. Prove that your answer is correct.
5. Let $g\colon \mathbb{R} \to \mathbb{Z}$ be the **ceiling function** $g(x) = \lceil x \rceil$, defined to be the smallest integer greater than or equal to x (i.e., "round x up to the nearest integer"; so $g(1.3) = 2$, and $g(3) = 3$.
   Compute the image of g. Prove that your answer is correct.
6. Consider the function $\sin\colon \mathbf{R} \to \mathbf{R}$.
   (a) Compute the image of sin. Prove that your answer is correct.
   (b) Compute $\sin([0, \pi/4])$. Prove that your answer is correct.
7. Let A and B be sets and let X and Y be subsets of A. Let $f\colon A \to B$ be a function. Prove or disprove each of the following. When giving a disproof, please give a counterexample.
   (a) $f(X \cup Y) \subseteq f(X) \cup f(Y)$.
   (b) $f(X \cup Y) \supseteq f(X) \cup f(Y)$.
8. Let A and B be sets and let X and Y be subsets of A. Let $f\colon A \to B$ be a function. Prove or disprove each of the following. When giving a disproof, please give an counterexample.
   (a) $f(X) - f(Y) \subseteq f(X - Y)$.
   (b) $f(X) - f(Y) \supseteq f(X - Y)$.

Recall that the homework is graded only for completion. If there are particular problems you would like grader feedback on, ~~please list them here.~~ Put a star by your problem number. (E.g., if you want feedback on problem 3, put a star by the number 3 in your solution.)

Assignment 9: Inverse Image (or "Preimage") and Injectivity

Due by 10:00am, eastern, on Friday, November 6

Suggested readings for this problem set: **Section 3.1, p. 90-93,**
All readings are from Bonde and Keane, *Introduction to Abstract Mathematics*.

Assignment: due Friday, November 6, 10:00am, via Gradescope (8D24GR):

(REMINDER: you are allowed to use the results of previous problem as part of the proof of later problems.)

1. Let $f\colon \mathbf{R} \to \mathbf{R}$ be the function defined by $f(x) = 3x + 1$.
   (a) Compute $f^{-1}(\{1, 5, 8\})$ (do not give a proof).
   (b) Compute $f^{-1}(W)$, where $W = (4, \infty)$, and give a proof that your answer is correct.
   (c) Compute $f^{-1}(\mathbf{E})$, where $\mathbf{E}$ is the set of even integers, and give a proof that your answer is correct. (Hint: note that $f(1/3)$ is even.)

2. Let $f\colon \mathbf{Z} \to \mathbf{Z}$ be the function defined by $f(n) = \begin{cases} \frac{n}{2}, & \text{if } n \text{ is even} \\ 2n + 4, & \text{if } n \text{ is odd.} \end{cases}$
   Compute $f^{-1}(\mathbf{E})$. Prove that your answer is correct. (Reminder: $\mathbf{E}$ is the set of even integers.)

3. Let A and B be sets and let X and Y be subsets of B. Let $f\colon A \to B$ be a function. Prove or disprove the following. (For a disproof, please give an explicit counterexample.)
   (a) $f^{-1}(X \cap Y) \subseteq f^{-1}(X) \cap f^{-1}(Y)$.
   (b) $f^{-1}(X \cap Y) \supseteq f^{-1}(X) \cap f^{-1}(Y)$.

4. Let A and B be sets and let X be a subset of B. Let $f\colon A \to B$ be a function. Prove or disprove the following. (For a disproof, please give an explicit counterexample.)
   (a) $X \subseteq f(f^{-1}(X))$.
   (b) $X \supseteq f(f^{-1}(X))$.

5. Let A and B be sets. Let $S \subseteq A$ and let $T \subseteq B$. Let $f\colon A \to B$ be a function. Prove or disprove the following. (For a disproof, please give an explicit counterexample.)
   (a) $f(S) \subseteq T \Rightarrow S \subseteq f^{-1}(T)$.
   (b) $S \subseteq f^{-1}(T) \Rightarrow f(S) \subseteq T$.

In each of the following problems, let $f\colon A \to A$ be a function (note that the domain and codomain are the same) and suppose that $C \subseteq A$.

6. Prove or disprove the following. (For a disproof, please give an explicit counterexample).
   (a) $f^{-1}(C) \subseteq C$;
   (b) $C \subseteq f^{-1}(C)$;
   (c) $f(C) \subseteq C$;
   (d) $C \subseteq f(C)$;
7. Prove or disprove the following (for a disproof, please give an explicit counterexample): $C \subseteq f^{-1}(C) \iff f(C) \subseteq C$.
8. Prove or disprove the following (for a disproof, please give an explicit counterexample): $f^{-1}(C) \subseteq C \iff C \subseteq f(C)$.

Recall that the homework is graded only for completion. If there are particular problems you would like grader feedback on, ~~please list them here.~~ Put a star by your problem number. (E.g., if you want feedback on problem 3, put a star by the number 3 in your solution.)

IN PROGRESS! Check back later for the final assignment.

Midterm 2 study guide

In class on Wednesday, November 11

Content: The questions will all be either

1. homework problems,
2. suggested problems,
3. problems we worked in class, or
4. minor variations of one of these.

Problems with very long proofs or that involved some unusual trick will not be on the exam.

You are allowed to use any previous problem from class or from the homework (e.g., "$A \subseteq A \cup B$") on the exam without reproving it, unless otherwise noted on the exam. (E.g., if I ask you to prove "$A \subseteq A \cup B$" on the exam, you will need to prove this using only the definition of subset and union, and I will remind you of this in the statement of the problem.)

A typical exam will have one or two questions from each week of the course. You can expect problems about following:

- Give definitions (e.g., subset, union, intersection, preimage, image)
- Proofs about sets (including power sets)
- Proofs about functions, images, preimages, surjectivity, injectivity

There will be at least one definition, and around 4-5 problems involving proofs (possibly including "prove or disprove" problems).

For definitions, I want a definition, in prose (complete sentences), and I want "just" the definition, and not any additional facts about the definition. (E.g., if you give the definition of rational, do not include that a rational number can be written in reduced form; that is a fact about rational numbers not part of the definition of rational.)

IN PROGRESS! Check back later for the final assignment.

IN PROGRESS! Check back later for the final assignment.

Assignment 10: More injectivity; composition of functions

Due by 10:00am, eastern, on Tuesday, 10

Suggested readings for this problem set: **Section 3.2, p. 100-105, Section 3.3, p. 110-113**
All readings are from Bonde and Keane, *Introduction to Abstract Mathematics*.

Assignment: due Friday, November 20, 10:00am, via Gradescope (8D24GR):

1. Let A and B be sets and let X and Y be subsets of B. Let $f\colon A \to B$ be a function. Prove or disprove the following. (For a disproof, please give an explicit counterexample.)
   (a) $f^{-1}(X - Y) \subseteq f^{-1}(X) - f^{-1}(Y)$.
   (b) $f^{-1}(X - Y) \supseteq f^{-1}(X) - f^{-1}(Y)$.
2. Let $f\colon A \to B$ be a function. Which of the followings statements are equivalent to the statement 'f is injective'? (No proof necessary.)
   (a) $f(a) = f(b)$ if $a = b$;
   (b) $f(a) = f(b)$ and $a = b$ for all $a, b \in A$;
   (c) If a and b are in A and $f(a) = f(b)$, then $a = b$;
   (d) If a and b are in A and $a = b$, then $f(a) = f(b)$;
   (e) If a and b are in A and $f(a) \neq f(b)$, then $a \neq b$;
   (f) If a and b are in A and $a \neq b$, then $f(a) \neq f(b)$.
3. Prove that the following functions are not injective.
   (a) $f\colon \mathbf{R} \to \mathbf{R}$, $f(x) = x^4 + x^2$;
   (b) $f\colon \mathbf{R} \to \mathbf{R}$, $f(x) = x^3 + x^2$;
   (c) $f\colon P(\mathbf{Z}) \to P(\mathbf{Z})$; $f(S) = S \cap \{1, 2\}$.
4. Prove that the following functions are injective.
   (a) $f\colon \mathbf{R}^2 \to \mathbf{R}^3$; $f(x, y) = (x + y, x - y, x^2 + y^2)$.
   (b) $f\colon \mathbf{R} \to \mathbf{R}$; $f(x) = e^{x+1}$.
   (c) $f\colon \mathbf{R} \to \mathbf{R}$, $f(x) = \begin{cases} -x - 1, & \text{if } x > 0 \\ x^2, & \text{if } x \leq 0. \end{cases}$

For problems 5, 6, and 7, let A, B and C be sets and let $f\colon A \to B$ and $g\colon B \to C$ be functions.

When giving a disproof, **please give a counterexample** (otherwise no credit will be given for the problem).

5. Prove or disprove: If $g \circ f$ is an injection, then g is an injection.

6. Prove or disprove each of the following.

   (a) If $g \circ f$ is a surjection, then f is a surjection.

   (b) If $g \circ f$ is a surjection, then g is a surjection.

7. Let A and B be sets and let $f\colon A \to B$ and $g\colon B \to A$ be functions. Prove that if $g \circ f$ and $f \circ g$ are bijective, then so are f and g.

8. We say that a function $f\colon \mathbf{R} \to \mathbf{R}$ is **increasing** if for all $x_1, x_2 \in \mathbf{R}$, if $x_1 < x_2$, then $f(x_1) < f(x_2)$.

   We say that a function $f\colon \mathbf{R} \to \mathbf{R}$ is **decreasing** if for all $x_1, x_2 \in \mathbf{R}$, if $x_1 < x_2$, then $f(x_1) > f(x_2)$.

   Let $f\colon \mathbb{R} \to \mathbb{R}$ and $g\colon \mathbb{R} \to \mathbb{R}$ be functions. Suppose that f and g are both decreasing. Prove that $g \circ f$ is increasing.

Recall that the homework is graded only for completion. If there are particular problems you would like grader feedback on, ~~please list them here.~~ Put a star by your problem number. (E.g., if you want feedback on problem 3, put a star by the number 3 in your solution.)

 IN PROGRESS! Check back later for the final assignment. 

Assignment 11: Inverse functions

Due by 10:00am, eastern, on Tuesday, December 4

Suggested readings for this problem set: **Section 3.3, p. 114-116**
All readings are from Bonde and Keane, *Introduction to Abstract Mathematics*.

Assignment: due Friday, December 4, 10:00am, via Gradescope (8D24GR):

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1. Define $f\colon \mathbf{R} - \{1\} \to \mathbf{R} - \{1\}$ by $f(x) = \frac{x+1}{x-1}$. Prove that f is a bijection. Find a formula for the inverse $f^{-1}(x)$, and prove that it is correct.
2. Let $f\colon \mathbf{R} \to \mathbf{R}$ be the function $f(x) = x^3 + x$. Prove that f is invertible without finding a formula for f^{-1}.
3. Let A, B and C be sets and let $f\colon A \to B$ and $g\colon B \to C$ be functions. Prove that if f and g are invertible, then so is $g \circ f$, and prove that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
4. Let A and B be sets and let $f\colon A \to B$ be a function. Suppose that f has a *left inverse* g; that is, suppose that there exists a function $g\colon B \to A$ such that $g \circ f = id_A$. Prove that f is injective.
5. Let $f\colon A \to B$ be a bijection. Let $g\colon B \to A$ be the inverse of f. Prove that g is also a bijection.
6. Let A be a set and let $f\colon A \to A$ be a function. Prove that $f \circ f = id_A$ if and only if "f is invertible and $f = f^{-1}$".
7. Let $f, g\colon A \to B$ be two functions, and let $h\colon B \to C$ be a function.
   (a) Prove that if h is injective and $h \circ f = h \circ g$ then $f = g$.
   (b) Give an example where h is not injective and $h \circ f = h \circ g$ but $f \neq g$.
8. Let $f\colon A \to B$ be a function, and let $g, h\colon B \to C$ be two functions.
   (a) Prove that if f is surjective and $g \circ f = h \circ f$ then $g = h$.
   (b) Give an example where f is not surjective and $g \circ f = h \circ f$ but $g \neq h$.

Recall that the homework is graded only for completion. If there are particular problems you would like grader feedback on, ~~please list them here.~~ Put a star by your problem number. (E.g., if you want feedback on problem 3, put a star by the number 3 in your solution.)

IN PROGRESS! Check back later for the final assignment.

IN PROGRESS! Check back later for the final assignment.

Assignment 12: Relations; (Un)countability

Due by 10:00am, eastern, on Tuesday, December 4

Suggested readings for this problem set: **Section 3.3, p. 114-116**
All readings are from Bonde and Keane, *Introduction to Abstract Mathematics*.

Assignment: due Friday, December 4, 10:00am, via Gradescope (8D24GR):

1. Define a relation R on the set $\mathbb{R}$ as follows: we say that $x \sim y$ if $|x| = |y|$. (Recall that $|x|$ is the absolute value of x) Determine whether this relation is reflexive, symmetric, transitive (and in each case give a proof or disproof).
2. Let $A = \{1, 2, 3\}$ and define a relation on A by $a \sim b$ if $a + b \neq 3$. Determine whether this relation is reflexive, symmetric, transitive (and in each case give a proof or disproof).
3. Define a relation on $\mathbf{Z}$ given by $a \sim b$ if $a - b$ is divisible by 3.
   (a) Prove that this is an equivalence relation.
   (b) What integers are in the equivalence class of 18? (No proof necessary.)
   (c) What integers are in the equivalence class of 31? (No proof necessary.)
   (d) How many distinct equivalence classes are there? What are they? (No proof necessary.)
4. Define a relation on $\mathbf{Z}$ given by $a \sim b$ if $a^2 - b^2$ is divisible by 4.
   (a) Prove that this is an equivalence relation.
   (b) How many distinct equivalence classes are there? What are they? (No proof necessary.)
5. Let A be a set, and let $P(A)$ be the power set of A. Assume that A is not the empty set. Define a relation on $P(A)$ by $X \sim Y$ if $X \subseteq Y$. Is this relation reflexive, symmetric, and/or transitive? In each case give a proof, or disprove with a counterexample. (For a counterexample, give an example of A, X, and Y that disproves the statement.)
6. Define a relation R on the set $\mathbb{R}$ as follows: we say that $x \sim y$ if $x - y \in \mathbb{Q}$. Determine whether this relation is reflexive, symmetric, transitive (and in each case give a proof or disproof).
7. Define a relation R on the set $\mathbb{Z}$ as follows: we say that $x \sim y$ if there exists a non-negative integer n such that $x = 2^n y$. Determine whether this relation is reflexive, symmetric, transitive (and in each case give a proof or disproof).
8. Define a relation R on the set $\mathbb{Z} \times (\mathbb{Z} - \{0\})$ as follows: we say that $(a, b) \sim (c, d)$ if $ad = bc$.
   (a) Prove that this is an equivalence relation.
   (b) What pairs (c, d) are in the equivalence class of $(1, 1)$? (No proof necessary.)
   (c) What pairs (c, d) are in the equivalence class of $(1, 2)$? (No proof necessary.)
   (d) How many distinct equivalence classes are there? What are they? (No proof necessary.)

1. Let S be one of the following sets. Give an example of a bijection from S to $\mathbb{Z}$. (No proof is necessary.)
   (a) $S = n\mathbb{Z}$ (for some fixed positive integer n).
   (b) $S = \mathbb{Z} \times \{0\}$.
   (c) $S = \mathbb{Z} \times \{-1, 1\}$.
   (d) $S = \sin^{-1}(\{0\})$.
2. Let $g\colon \mathrm{Fun}(S, \{0,1\}) \to P(S)$ be the function $f \mapsto f^{-1}(\{0\})$. Prove that g is a bijection.

3. As in class, let $P_{\mathrm{BD}}(S)$ be the set of all finite subsets of S, i.e.,
$$P_{\mathrm{BD}}(S) = \{A \subset S \mid |A| < \infty\}$$
and for $i \in \mathbb{Z}_{>0}$ let
$$P_i(S) = \{A \subset S \mid |A| \leq i\}.$$
   (a) Prove that
$$P_{\mathrm{BD}}(S) = \bigcup_{i \in \mathbb{Z}_{>0}} P_i(S).$$
   (b) Prove that the map
$$S^i \to P_i(S)$$
   given by
$$(a_1, \ldots, a_i) \mapsto \{a_1, \ldots, a_i\}$$
   is a surjection.
   (c) Prove that $P_{\mathrm{BD}}(\mathbb{Z})$ is countable.

4. Prove that $|S| \neq |\mathrm{Fun}(S, \{0,1\})|$ by the same technique as the proof from class that $|S| \neq |P(S)|$. (In other words, proceed by contradiction, assuming that there is a bijection $f\colon S \to \mathrm{Fun}(S, \{0,1\})$ and then construct some $g \in \mathrm{Fun}(S, \{0,1\})$ which is not in the image of f.)

Recall that the homework is graded only for completion. If there are particular problems you would like grader feedback on, ~~please list them here.~~ Put a star by your problem number. (E.g., if you want feedback on problem 3, put a star by the number 3 in your solution.)

IN PROGRESS! Check back later for the final assignment.

IN PROGRESS! Check back later for the final assignment.

Final exam study guide

Final exam is TBA

The exam will be available on gradescope, and should be submitted via Gradescope.

Big request: if you are writing your exam in latex or on a tablet, please start each problem on a new page. (My eyes are not very good and this helps a lot when I am grading ≥ 50 exams.)

The **last day of class** is Wednesday, December 9.

The *types* of problems will include a subset of

1. Computations
2. Proofs
3. Algorithms
4. True False
5. Bonus problem

Problems with extremely long proofs or that involved some unusual trick will not be on the exam.

Since this is a take home exam, none of the problems will be identical to homework problems, but many problems will be minor variations of homework or of problems we worked in class.

A good way to prepare is to:

1. Know all of the definitions and terminology;
2. Know all of the statements of theorems, and examples of how we use the theorems;
3. Make a list of all of the different *proof techniques* from class and from the homework and review how those techniques are used in proofs and problems;
4. Practice doing problems "from scratch" and use your solutions as "hints" when you get stuck.

Additionally:

1. You are allowed to use the textbook, lecture notes and any materials from the course website.
2. Using Google or any other online resources is not always a reliable source. Please do not use Chat GPT or any other AI assistants.
3. You are allowed to use Theorems, lemmas, etc from the book or from class as part of your solutions, and you are not required to reprove these during the exam. Please do cite them (e.g., "Proposition 1.3.4 from our book") or refer to them by name, if they have a special name (e.g., "by the Matrix Tree Theorem").

4. Do not discuss the problems or their solutions with your classmates.

5. You can always ask me (the instructor) if you have clarifying questions, but asking for hints or asking if a proof is correct is not allowed.

The exam will be **comprehensive**. A typical exam will have a few questions from each week of the course, and there will be more emphasis on content not covered on the midterms. The exam to be around 8-9 problems (some with multiple parts), and 2-3 bonus questions.